Stock Price Prediction Using News Sentiment

Andre Barle, Harsha, Pranay

Problem Statement - Motivation

Problem Statement

- Stock prices are influenced by news and public sentiment, yet most predictive models focus on quantitative data and overlook qualitative factors
- The goal is to predict whether a stock's price will go up or down based on daily news sentiment and historical stock data
- Use machine learning models and reinforcement learning to simulate and optimize trading decisions

Motivation

- News directly affects market behavior, as headlines can shift investor mood in seconds
- Most trading models rely on technical indicators, as they miss the "why" behind movements
- Advances in NLP allow us to accurately measure sentiment and combine it with price data for more informed, profitable decision

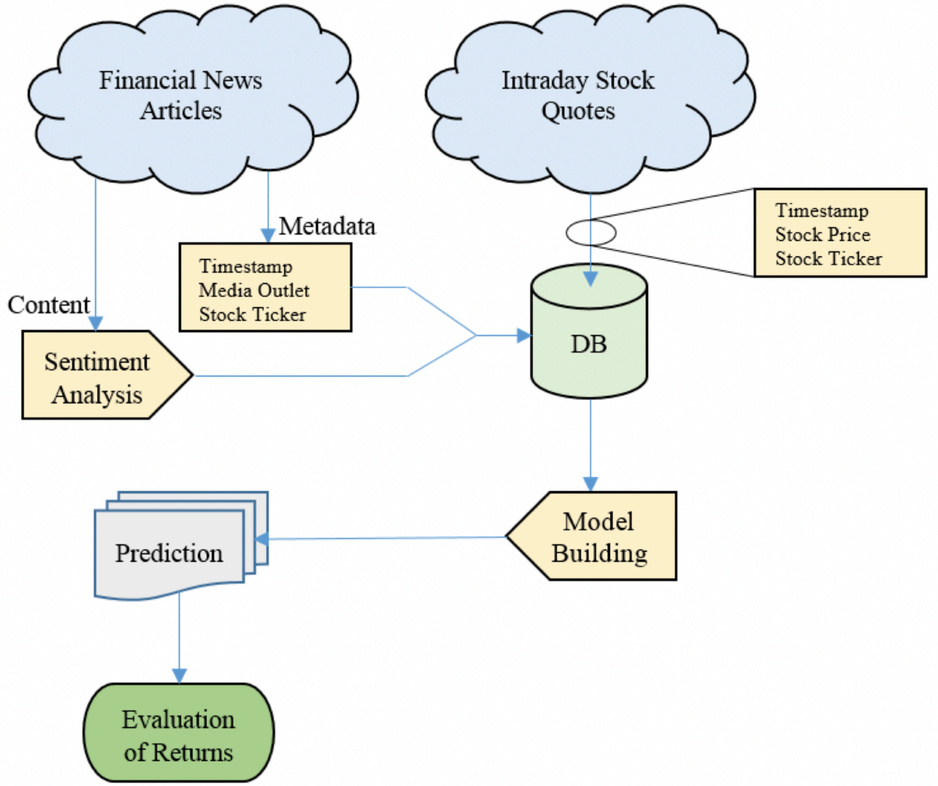


Figure 1. CentralFinance System

Schumaker, Robert P. and Maida, Nick (2018) "Analysis of Stock Price Movement Following Financial News Article Release," *Communications of the IIMA*: Vol. 16: Iss. 1, Article 1.
DOI: https://doi.org/10.58729/1941-6687.1378

Prior Work Review

- P. V. Chandrika, K. Visalakshmi and K. Sakthi Srinivasan, "Application of Hidden Markov Models in Stock Trading," 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2020, pp. 1144-1147, doi: 10.1109/ICACCS48705.2020.9074387.

- A. Gupta and B. Dhingra, "Stock market prediction using Hidden Markov Models," 2012 Students Conference on Engineering and Systems, Allahabad, India, 2012, pp. 1-4, doi: 10.1109/SCES.2012.6199099.

- Nguyen, N.; Nguyen, D. Hidden Markov Model for Stock Selection. Risks 2015, 3, 455-473. https://doi.org/10.3390/risks3040455

- Kim, J.; Kim, H.-S.; Choi, S.-Y. Forecasting the S&P 500 Index Using Mathematical-Based Sentiment Analysis and Deep Learning Models: A FinBERT Transformer Model and LSTM. Axioms 2023, 12, 835. https://doi.org/10.3390/axioms12090835

Contributions

Harsha – **Data Preprocessing & Integration:**

- Cleaned and filtered the original dataset containing stock tickers, titles and dates
- Integrated stock prices by referencing external CSVs using ticker and date
- Handled formatting, missing values, and consistency for for sentiment labeling and model training.

Pranay – **Cleaning & Scoring Article Headlines**

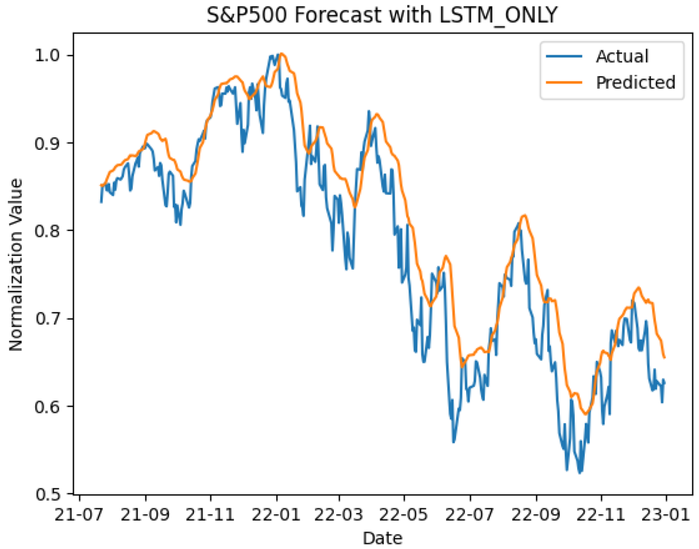
- Added a news sentiment plugin that reads each headline and rates it from negative to positive
- Processed all the headlines, removed extreme outliers, and rescaled the scores to a simple -1 to +1 range

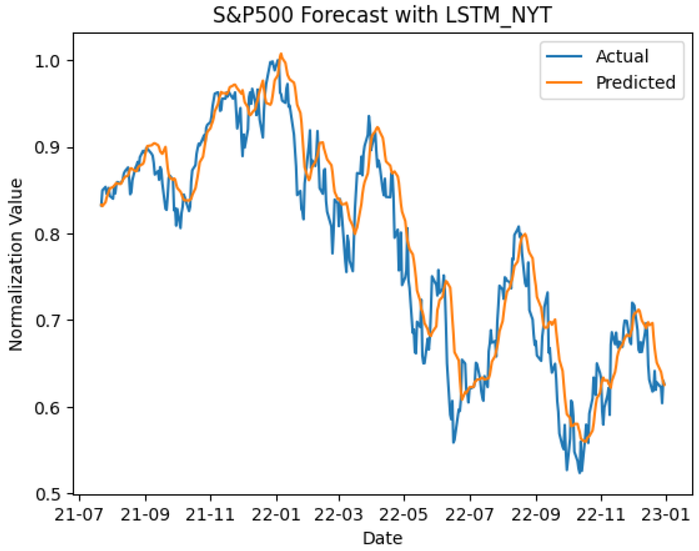
Andre - **Hidden Markov Model with Logistic Regression and PPO Agent**

- Trained Hidden Markov Model to be fed into Logistic Regression Classifier Model for stock price change prediction
- PPO Agent to make decision on how to best trade in a given environment, essentially acting as a stock picker

Proposed Method

- **Target Labeling**: Compute each headline's next-day price change and convert it to a binary "up" (1) vs. "down" (0) label.
- **Feature Engineering**:
  - Scale raw FinBERT sentiment scores into a –1…+1 range
  - Map each stock ticker to a numeric ID
  - Fit a 2-state Gaussian HMM on these features to capture latent market regimes
- **Predictive Modeling**: Split data at 2017-01-01 (UTC), train a Logistic Regression on [sentiment, ticker_id, HMM_state], and validate to achieve ~70% test accuracy.
- **Trading Environment**: Build a custom Gym TradingEnv with four discrete actions (flatten, hold, long all, short all), track cash/holdings, and define reward as the change in portfolio value.
- **Reinforcement-Learning Optimization**: Vectorize the environment, train multiple PPO agents over the test series, and select the agent that yields the highest final portfolio value as the optimized strategy.





Kim, J.; Kim, H.-S.; Choi, S.-Y. Forecasting the S&P 500 Index Using Mathematical-Based Sentiment Analysis and Deep Learning Models: A FinBERT Transformer Model and LSTM. *Axioms* **2023**, *12*, 835. https://doi.org/10.3390/axioms12090835

Dataset

Original Dataset (from Kaggle):

- Source: Daily Financial News for 6000+
- Dataset Fields:
  - Ticker, Date, News Title

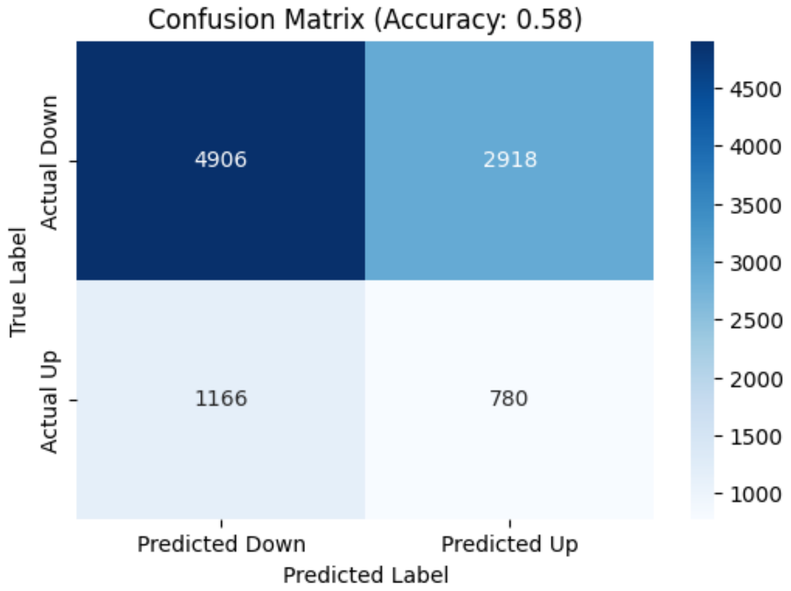
Modified Dataset (our version):

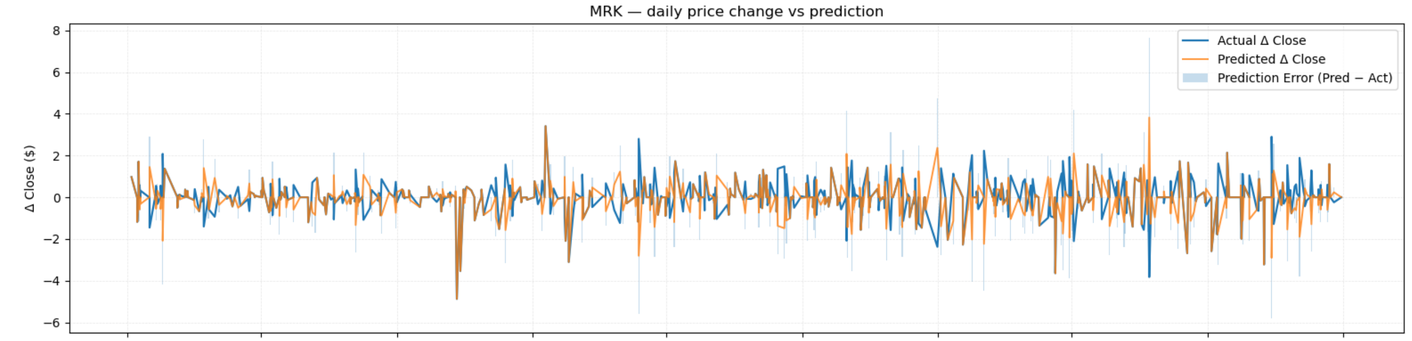
- Added stock price data from external CSVs matched by ticker and date
- Added, sentiment scores for each news title using FinBERT
- Final Dataset Fields:
  - Ticker, Date, News Title, Open/Close prices, Sentiment

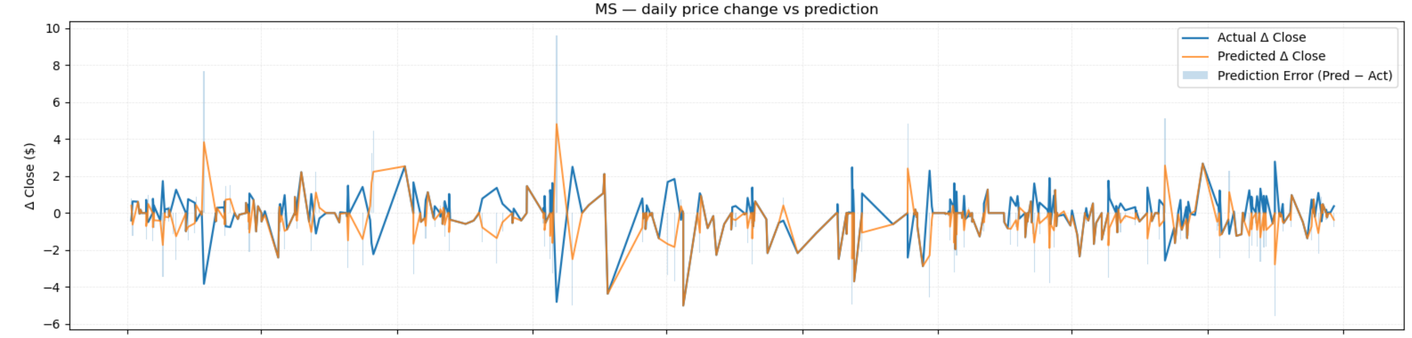
	Unnamed: 0	title	date	stock
0	0.0	Stocks That Hit 52-Week Highs On Friday	2020-06-05 10:30:00-04:00	A
1	1.0	Stocks That Hit 52-Week Highs On Wednesday	2020-06-03 10:45:00-04:00	A
2	2.0	71 Biggest Movers From Friday	2020-05-26 04:30:00-04:00	A
3	3.0	46 Stocks Moving In Friday's Mid-Day Session	2020-05-22 12:45:00-04:00	A
4	4.0	B of A Securities Maintains Neutral on Agilent...	2020-05-22 11:38:00-04:00	A
...	...	...	...	...
1400464	1413844.0	Top Narrow Based Indexes For August 29	2011-08-29 10:41:00-04:00	ZX
1400465	1413845.0	Recap: Wednesday's Top Percentage Gainers and ...	2011-06-22 16:44:00-04:00	ZX
1400466	1413846.0	UPDATE: Oppenheimer Color on China Zenix Auto ...	2011-06-21 08:26:00-04:00	ZX
1400467	1413847.0	Oppenheimer Initiates China Zenix At Outperfor...	2011-06-21 05:59:00-04:00	ZX
1400468	1413848.0	China Zenix Auto International Opens For Tradi...	2011-05-12 09:36:00-04:00	ZX

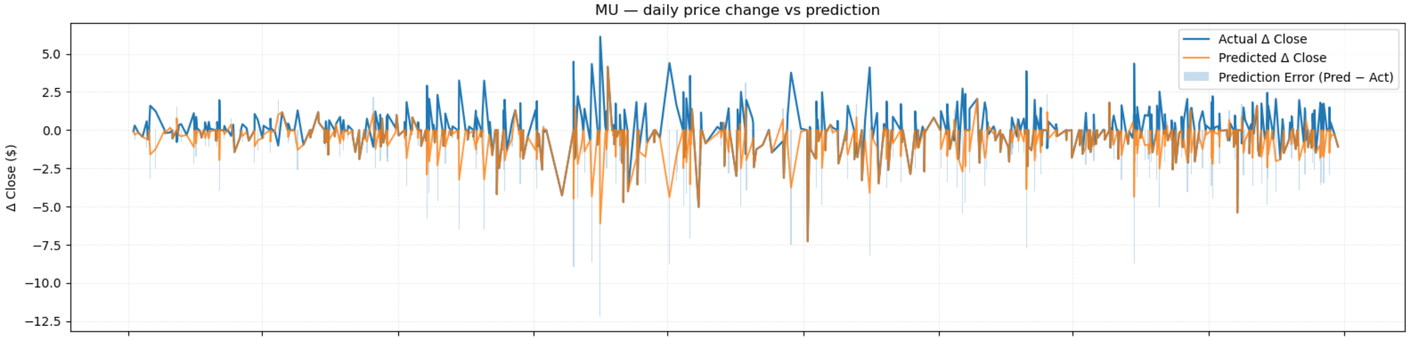
1400469 rows × 4 columns

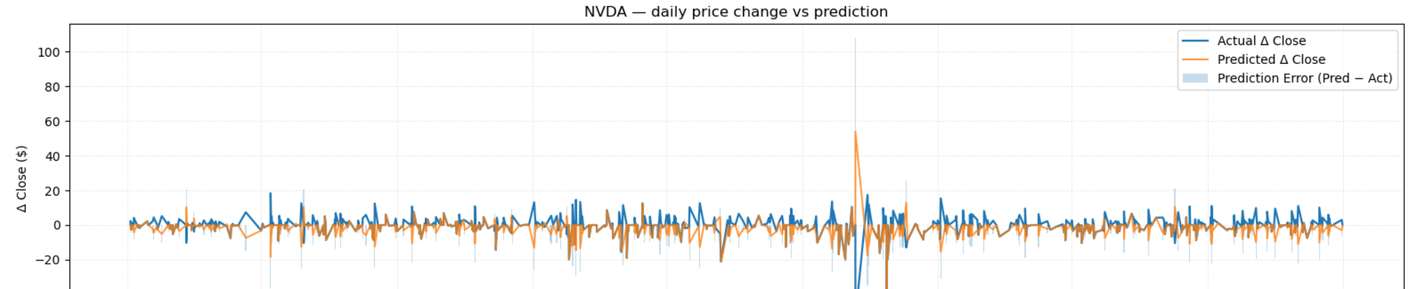
Results


Confusion Matrix (Accuracy: 0.58)
True Label
Actual Down
Actual Up
4906
2918
1166
780
Predicted Down
Predicted Up
Predicted Label
4500
4000
3500
3000
2500
2000
1500
1000

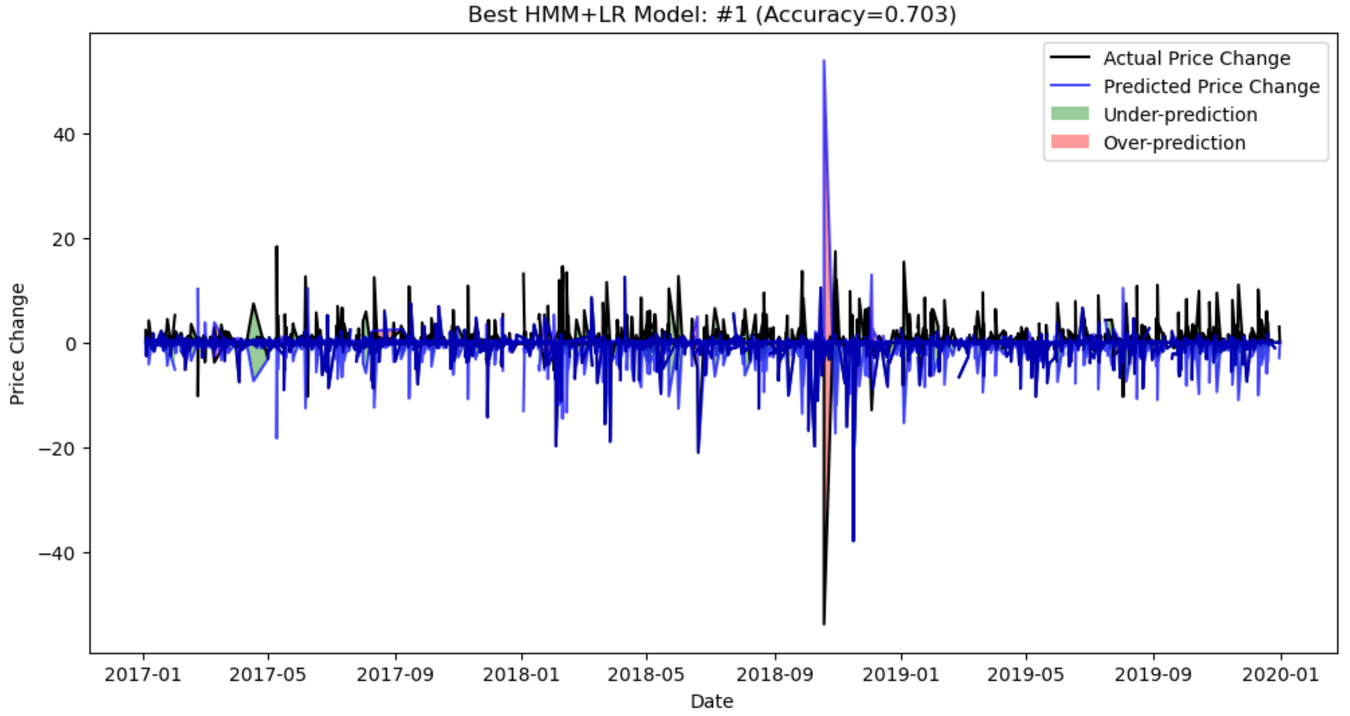

MRK — daily price change vs prediction
Actual Δ Close
Predicted Δ Close
Prediction Error (Pred − Act)
Δ Close ($)

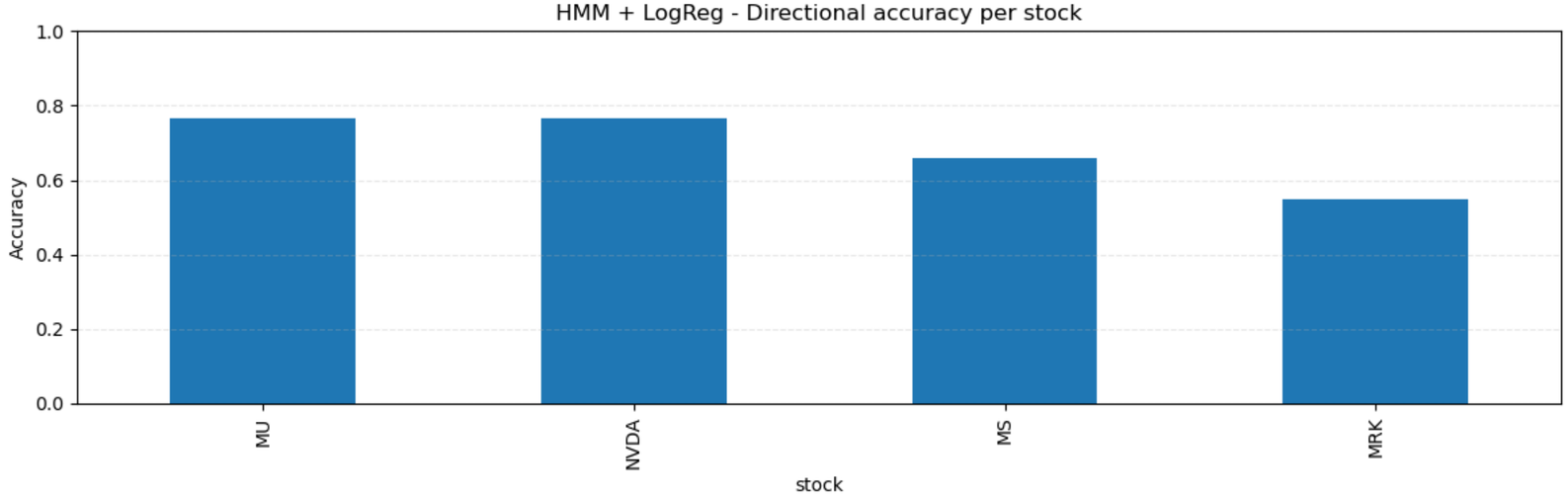

MS — daily price change vs prediction
Actual Δ Close
Predicted Δ Close
Prediction Error (Pred − Act)
Δ Close ($)


MU — daily price change vs prediction
Actual Δ Close
Predicted Δ Close
Prediction Error (Pred − Act)
Δ Close ($)


NVDA — daily price change vs prediction
Actual Δ Close
Predicted Δ Close
Prediction Error (Pred − Act)
Δ Close ($)

Results


Best HMM+LR Model: #1 (Accuracy=0.703)
Actual Price Change
Predicted Price Change
Under-prediction
Over-prediction
Price Change
40
20
0
−20
−40
2017-01
2017-05
2017-09
2018-01
2018-05
2018-09
2019-01
2019-05
2019-09
2020-01
Date


HMM + LogReg - Directional accuracy per stock
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
MU
NVDA
MS
MRK
stock

Discussion

- Hitting ~70% accuracy shows our scaled FinBERT scores really do capture market mood for next-day move although because of the noise in the data, there have been a few incorrect scores.
- Our PPO agent decides each morning whether to flatten, hold, buy, or short, and in backtests it beats just buying and holding.
- By folding in the HMM's hidden-state labels, the agent recognizes whether markets are "calm" or "volatile" and changes its aggressiveness accordingly.
- From pulling headlines and scoring sentiment to running through portfolio simulations, we've built a seamless flow that shows news-driven algo trading is practically achievable.
- Future work: This can be complemented with a live setup of buying and selling stocks. In a live environment, news headlines are received and scored in real time, tagged with a market sentiment, then fed to the PPO policy to generate buy/hold/sell signals, possibly via a broker API with automated position sizing, stop-loss management, and daily model retraining for continuous adaptation.

Questions